Value putting:

i) No. of variables > no. of eqns.

ii) Formula asked $\begin{cases} v \to v \\ v \to c \end{cases}$

$\theta \to 90°/0°,\ 45°,\ 30°/60°$

$\frac{\theta}{2}, 2\theta \to 60, 30, 15$

iii) Misc.
- Putting all angles equal. (Symmetrical)
- Substituting complementary values.

1. The simplified value of $(secA - cosA)^2 + (cosecA - sinA)^2 - (cotA - tanA)^2$

$(secA - cosA)^2 + (cosecA - sinA)^2 - (cotA - tanA)^2$ को सरल करें:

a) 0 b) $\frac{1}{2}$

c) 1 d) 2

2. **The value of $(1 + cot\theta - cosec\theta)(1 + tan\theta + sec\theta)$ is equal to**

a) 1 **b) 2**

c) 0 **d) -1**

(3) If $(sin\alpha + cosec\alpha)^2 + (cos\alpha + sec\alpha)^2 = k + \tan^2\alpha + \cot^2\alpha$, then the value of k is

अगर $(sin\alpha + cosec\alpha)^2 + (cos\alpha + sec\alpha)^2 = k + tan^2\alpha + cot^2\alpha$, है, तो k का मान:

a) 1 b) 7

c) 3 d) 5

(4) Find the value of $3\left(\sin^4\theta+\cos^4\theta\right)+2\left(\sin^6\theta+\cos^6\theta\right)+12\sin^2\theta\cos^2\theta$

$3\left(\sin^4\theta+\cos^4\theta\right)+2\left(\sin^6\theta+\cos^6\theta\right)+$

$2 \qquad 2$

a) 3 b) 2

c) 0 d) 5

(5) The simplified value of $(sec\ x\ sec\ y + tan\ x\ tan\ y)^2 - (sec\ x\ tan\ y + tan\ x\ sec\ y)^2$ $(sec\ x\ sec\ y + tan\ x\ tan\ y)^2 - (sec\ x\ tan\ y + tan\ x\ sec\ y)^2$ को सरल करें|

a) -1 b) 0

c) $\sec^2 x$ d) 1

$$\frac{-\sin^2}{-\sin^2} = \frac{}{\cos^2 y - 1} = \frac{-\sin^2 y}{(1+\cos y)(\cos y - 1)}$$

$x = 0$

(6) What is the value of $\frac{[(\sin x+\sin y)(\sin x-\sin y)]}{(\cos x+\cos y)(\cos y-\cos x)}$?

$\frac{[\sin x+\sin y \quad \sin x-\sin y]}{\cos x+\cos y \quad \cos y-\cos x}$ का मान क्या है?

a) 0 b) 1

c) -1 d) 2

$y = 0$

$$\frac{\sin^2 x}{(\cos x+1)(1-\cos x)} = \frac{\sin^2}{\sin^2}$$

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7 What is the value of $\sin(B-C)\cos(A-D)+\sin(A-B)\cos(C-D)+\sin(C-A)\cos(B-D)$?

$\sin B-C \cos A-D +\sin A-B \cos(C-D)+\sin C-A \cos B-D$ का मान क्या है?

a) $\frac{3}{2}$ b) -3

c) 1 d) 0

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⑧ What is the value of $cos(90-B)\,sin(C-A)+sin(90+A)\,cos(B+C)-sin(90-C)\,cos(A+B)$?
$cos\ 90-B\ \ sin\ C-A\ \ +sin\ 90+A\ \ cos(B+C)-sin(90-C)\,cos(A+B)$ का मान क्या है?

a) 1 b) $sin(A+B-C)$

c) 0 d) $cos(B+C-A)$

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⑨ **What is the value of**

$$\frac{[sin(y-z)+sin(y+z)+2sin\,y]}{sin(x-z)+sin(x+z)+2sin\,x}\ ?$$

$$\frac{[sin\ y-z\ +sin\ y+z\ +2sin\,y]}{sin\ x-z\ +sin\ x+z\ +2sin\,x}$$ **का मान क्या है?**

a) $cosx\ sin\,y$

b) $\frac{sin\,y}{sin\,x}$

c) $sin\,z$

d) $sinx\ tany$

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10. What is the value of $[(Cos^3 2\theta + 3\,Cos\,2\theta) \div (Cos^6\,\theta - Sin^6\theta)]$?

$[(\underset{6}{Cos^3}\ 2\theta + \underset{6}{3\,Cos}\ 2\theta) \div$

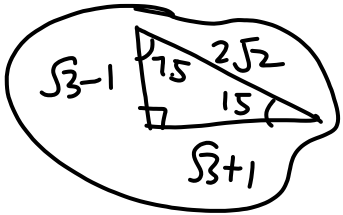
a) 0 b) 1
c) 4 d) 2

SSC CGL 2017 Mains

(11) $\frac{\cot\theta}{\cot\theta-\cot 3\theta}+\frac{\tan\theta}{\tan\theta-\tan 3\theta}=?$

a) 1 b) -1

c) 2 d) 0



(12) The value of $\frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1} \times \frac{\tan^2\theta(\text{cosec}^2\theta-1)}{\sec\theta-\tan\theta}$ is:

$\frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1} \times \frac{\tan^2\theta\ \text{cosec}^2\theta-1}{\sec\theta-\tan\theta}$ का मान क्या है?

a) 0 b) -1

c) 1 d) $\frac{1}{2}$

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(13) $\frac{(2sinA)(1+sinA)}{\frac{1+sinA+cosA}{2sinA \quad 1+sinA}}$ is equal to:

a) $1 + sinA - cosA$

b) $1 - sinAcosA$

c) $1 + cosA - sinA$

d) $1 + sinAcosA$

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(14) What is the value of $\sin(90° + 2A)[4 - \cos^2(90° - 2A)]$?
$\sin(90° + 2A)[4 - \cos^2(90° - 2A)]$ का मान क्या है?

a) $2(cos^3A - sin^3A)$
b) $2(cos^3A + sin^3A)$
c) $4(cos^6A + sin^6A)$
d) $4(cos^6A - sin^6A)$

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15

HW

If $x = cosec\theta - \sin\theta$ and $y = \sec\theta - \cos\theta$, then the relation between x and y is

यदि $x = cosec\theta - \sin\theta$ और $y = \sec\theta - \cos\theta$ हो, तो x और y के बीच सम्बन्ध होगा

a) $x^2 + y^2 + 3 = 1$

b) $x^2y^2(x^2 + y^2 + 3) = 1$

c) $x^2(x^2 + y^2 - 5) = 1$

d) $y^2(x^2 + y^2 - 5) = 1$

SSC CGL 2016 T-2

16 HW If $cot\,\theta + tan\,\theta = p$ and $sec\,\theta - cos\,\theta = q$ then $(p^2q)^{\frac{2}{3}} - (pq^2)^{\frac{2}{3}} = ?$

यदि $cot\,\theta + tan\,\theta = p$ और $sec\,\theta - cos\,\theta = q$ है, तो $p^2q^{\frac{2}{3}} - pq^{2\,\frac{2}{3}} = ?$

a) 0 b) 1

c) 2 d) 3

CDS 2016 I

(17) If $\frac{cos\alpha}{sin\alpha+cos\beta}+\frac{cos\beta}{sin\beta-cos\alpha}=\frac{x}{sin\alpha-cos\beta}+\frac{cos\beta}{sin\beta+cos\alpha}$ then 'x' is equal to:

यदि $\frac{cos\alpha}{sin\alpha+cos\beta}+\frac{cos\beta}{sin\beta-cos\alpha}=\frac{x}{sin\alpha-cos\beta}+\frac{cos\beta}{sin\beta+cos\alpha}$ है, 'x' तो बराबर है:

a) $cos\beta$ b) $cos\alpha$

c) $sin\beta$ d) $sin\alpha$

SSC CHSL 2018

(18) What is the value of $\left[(sec\,2\theta + 1)\sqrt{sec^2\theta - 1}\right] \times \frac{1}{2}(cot\ \theta - tan\ \theta)$?

$\left[(sec\,2\theta + 1)\sqrt{sec^2\theta - 1}\right] \times \frac{1}{2}(cot\,\theta - tan\,\theta)$ का मान क्या है?

a) 0 b) 1

c) $cosec\ \theta$ d) $sec\ \theta$

SSC CGL 2017 Mains

(19) If $sinA + sinB = C$ & $cosA + cosB = D$, then $sin(A + B)$ =?

यदि $sinA + sinB = C$ और $cosA + cosB = D$ है, तो $sin\ A + B$ =?

a) CD

b) $\frac{CD}{C^2+D^2}$

c) $\frac{C^2+D^2}{2CD}$

d) $\frac{2CD}{C^2+D^2}$

20 If $\alpha + \beta = 90°$ then the value of $(1 - \sin^2 \alpha)(1 - \cos^2 \alpha) \times (1 + \cot^2 \beta)(1 + \tan^2 \beta)$ is

अगर $\alpha + \beta = 90°$ है, तो $(1 - sin^2 \alpha)(1 - cos^2 \alpha) \times (1 + cot^2 \beta)(1 + tan^2 \beta)$ का मान:

a) 1 b) -1

c) 0 d) 2

21

HW

If $\angle A$ and $\angle B$ are complementary to each other, then the value of $\sec^2 A + \sec^2 B - \sec^2 A \sec^2 B$ is

अगर कोण $\angle A$ और $\angle B$ एक दुसरे के पूरक हैं, तो $sec^2 A + sec^2 B - sec^2 A sec^2 B = ?$

a) 1 b) -1

c) 2 d) 0

22 HW

The value of the expression $(cos^6\theta + sin^6\theta - 1)(tan^2\theta + cot^2\theta + 2)$ is:

व्यंजक $(cos^6\theta + sin^6\theta - 1)(tan^2\theta + cot^2\theta + 2)$ निम्न में से किसके बराबर है?

a) 0 b) -1

c) -3 d) 1

SSC CGL 2018 T-2

(23) $tan\left(\frac{\pi}{4}+\theta\right)-tan\left(\frac{\pi}{4}-\theta\right)=?$

a) $2\ tan\ 2\theta$ b) $2\ cot\ 2\theta$

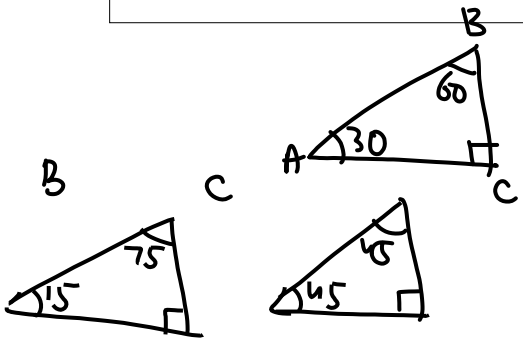
c) $tan\ 2\theta$ d) $cot\ 2\theta$

(24) If A, B, C are angles of a triangle, then $sin^2A + sin^2B + sin^2C - 2cosA\ cosB\ cosC = ?$

यदि A, B, C एक त्रिभुज के कोण हैं, तो $sin^2A + sin^2B + sin^2C - 2cosA\ cosB\ cosC = ?$

a) 1 b) 2

c) 3 d) 4



25 If $A + B + C = \frac{3\pi}{2}$, then $cos2A + cos2B + cos2C = ?$

यदि $A + B + C = \frac{3\pi}{2}$ है, तो $cos2A + cos2B + cos2C = ?$

a) $1 - 4\,sinA\,sinB\,sinC$

b) $1 - sinA\,sinB\,sinC$

c) $1 - 2sinA\,sinB\,sinC$

d) $1 - 4cosA\,cosB\,cosC$